Discrete Ricci Flow On Special Graph Structures

Andrew Ely

In this paper, we analyze the long-time behavior of specific, special graph structures under continuous-time Discrete Ricci Flow based on the Lin-Lu-Yau version of Ollivier Ricci curvature. The results prove convergence to limiting metrics of several specific graph structures with specific conditions under Discrete Ricci Flow.

Introduction

Richard Hamilton [1] introduced Ricci Flow in the 1980's, which evolves Riemannian metrics on manifolds. Grigori Perelman [2] presented a number of new results in 2002 and 2003 building on Hamilton's Ricci Flow, leading to the proof of the Poincaré conjecture. These works by Hamilton and Perelman are milestones in differential geometry and topology.

Discrete Ricci Flow is an adaptation of Riemannian Ricci Flow for discrete metric spaces. Discrete Ricci Flow is an iterative process that evolves a discrete metric space via a system of ODEs. There are various versions of Discrete Ricci Flow, such as Bakry-Emery [3], Forman [4], and Ollivier [5]. Lin-Lu-Yau [6] curvature is a further adaptation of Ollivier curvature, which has been used for analyzing Discrete Ricci Flow on the discrete metric space defined by graphs. The main application of Discrete Ricci Flow on graph networks is community detection in large networks [7]. Intuitively, the Discrete Ricci Flow process evolves a graph's edge weights via edge curvature, where negative curvature causes edge weights to stretch and positive curvature causes edge weights to shrink, exposing underlying community structures. Theoretical analysis of Discrete Ricci Flow is interested in answering what the underlying curvature of a given graph is, if the solution to the Ricci Flow always exists and at what domains, if the limiting metric exists with constant curvature, and if a graph converges to a limiting metric. In [8], Shuliang Bai et al. provide rigorous convergence results for tree graphs under Lin-Lu-Yau Ricci Flow. Trees have no cycles, which simplifies the underlying curvature and ODE system that dictates the flow. Cyclic graphs, however, result in systems that are much harder to analyze. Since there is no known variational structure for expressing the curvature of edges in cyclic graphs, and no known variational structure for a Ricci Flow, proving the convergence of Discrete Ricci Flow is a difficult task. In this paper, motivated by the authors and works of ([6],[7],[8],[9]), we provide proof of convergence of Lin-Lu-Yau curvature on specific cyclic graph structures, as well as one unique tree structure.

Preliminaries and Definitions

The following definitions are taken from [8] by Shuliang Bai et al.

First, we must define the metric space of a graph. A metric space is a pair (X, d) where X is a set and d is a distance function $d : X \times X \to \mathbb{R}_{\geq 0}$ with the properties: **(1)** $d(x, y) = 0$ if and only if $x = y$ for all $x, y \in X$. **(2)** $d(x, y) = d(y, x)$ for all $x, y \in X$. And **(3)** $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$. Let $G = (V, E, w)$ be a weighted graph with edge set E and vertex set V. The function $w((x, y)) : E \to [0, \infty)$ is the associated edge weight between vertices $(x, y) \in E$. We denote $x \sim y$ as vertices that have an edge weight between them, and $w_{xy}(t)$ as the edge weight at time t. A distance metric $d(x, y) : V \times V \to \mathbb{R}_{\geq 0}$ is defined by the length of a minimal weighted path among all paths that connect vertices x, y. We denote $d_{xy}(t)$ to be the distance metric between vertices $(x, y) \in V$ at time t. Equipped with an edge weight metric w being the set of edge weights with initial weights strictly positive, and distance metric d being the shortest distance between all pairs of vertices, a graph is a metric space defined by the pair (w, d). If there exists an edge $(x, y) \in E$ such that $d_{xy}(t) \neq w_{xy}(t)$, the distance condition is violated. In other words, it is critical for the expression of edge curvature that any vertices that have an edge weight between them have a distance metric equal to that edge weight. One remedy for the distance condition violation is to delete such an edge weight before the flow procedure continues. In this paper, the graphs we analyze have initial conditions and structures that prevent the distance violation from ever occurring. Under these conditions, by Theorem 3 in [8], there exists a unique solution for all time to the system of ordinary differential equations in normalized and unnormalized Discrete Ricci Flow. Next, we state all relevant definitions from [8] that we will use in the rest of the paper.

Definition 1. A probability distribution associated with a vertex x over all vertices $y \in V(G)$ is a mapping $\mu : V \to [0, 1]$ defined by

$$\mu_x^\alpha(y) = \begin{cases} \alpha & \text{if } y = x \\ (1-\alpha)\frac{\gamma(w_{x,y})}{\sum\limits_{z \sim x} \gamma(w_{x,z})} & \text{if } y \sim x \\ 0 & \text{otherwise} \end{cases}$$

Where $\gamma : \mathbb{R}_+ \to \mathbb{R}_+$ represents an arbitrary one-to-one function, and $\alpha \in [0, 1]$.

Definition 2. A coupling between two probability distributions μ_u and μ_v is a mapping $A : V \times V \to [0, 1]$ with finite support such that

$$\sum_{y \in V} A(x, y) = \mu_u(x) \quad \text{and} \quad \sum_{x \in V} A(x, y) = \mu_v(y)$$

Finite support means the distribution is nonzero on only finitely many vertices.

Definition 3. The transportation distance between two probability distributions μ_u and μ_v is given by the Wasserstein Function

$$W(\mu_u, \mu_v) = \inf_A \sum_{x,y \in V} A(x, y)d(x, y)$$

Where the infimum is taken over all couplings A between μ_u and μ_v, and $d(x, y)$ is the distance metric. The Wasserstein distance is also equal to the optimal solution to the dual problem

$$W(\mu_u, \mu_v) = \sup_f \sum_{x \in V} f(x)[\mu_u(x) - \mu_v(x)]$$

Where the supremum is taken over all 1-Lipschitz functions f. f is 1-Lipschitz if it satisfies

$$|f(x) - f(y)| \leq d(x, y) \quad \text{for } \forall x, y \in V(G)$$

Definition 4. Let $G = (V, E, w)$ be a weighted graph equipped with a distance metric d determined by the shortest weighted paths between nodes. For any $(x, y) \in E(G)$ and $\alpha \in [0, 1]$, the α-Ricci curvature k_α on edge (x, y) is defined to be

$$k_\alpha(x, y) = 1 - \frac{W(\mu_x^\alpha, \mu_y^\alpha)}{d(x, y)}$$

And the Lin-Lu-Yau version of Ollivier Ricci curvature is defined as

$$k(x, y) = \lim_{\alpha \to 1} \frac{k_\alpha(x, y)}{1 - \alpha}$$

Definition 5. Given an initial edge weight metric $w_{xy}(0)$ for each edge $(x, y) \in E(G)$, the system of ordinary differential equations given by

$$\frac{dw_{xy}(t)}{dt} = -k_{xy}(t)w_{xy}(t) \tag{1}$$

defines the unnormalized flow on graph G. The normalized flow, which ensures the sum of all edge weights remains constant throughout time is given by

$$\frac{dw_{xy}(t)}{dt} = -k_{xy}(t)w_{xy}(t) + w_{xy}(t) \sum_{e \in E(G)} k_e w_e(t) \tag{2}$$

Where if the initial edge weights sum to some constant 1 at $t = 0$, then for all time the sum will remain constant 1. The connection between normalized and unnormalized flow is given by

$$w_{xy}(t) = \frac{\hat{w}_{xy}(t)}{\sum_{e \in E} \hat{w}_e(t)} \tag{3}$$

Where $\hat{w}_e$ denotes unnormalized edge weight, and w_e denotes normalized edge weight.

Definition 6. Let $G = (V, E)$ be equipped with an initial metric $w \in \mathbb{R}^{|E|}_{>0}$ and let the metric $w(t)$ evolve under the normalized Ricci flow. We say that the normalized Ricci flow converges to a metric with constant curvature τ if there exists a metric $w(\infty) \in \mathbb{R}^{|E|}_{\geq 0}$ such that $\lim_{t \to \infty} w(t) = w(\infty)$ and under the metric $w(\infty)$ the curvature on each edge in $E^+ := \{e \in E(T) : w_e(\infty) > 0\}$ is τ.

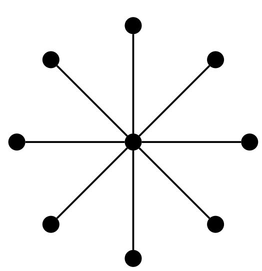
Main Results

Ricci Flow On Trees

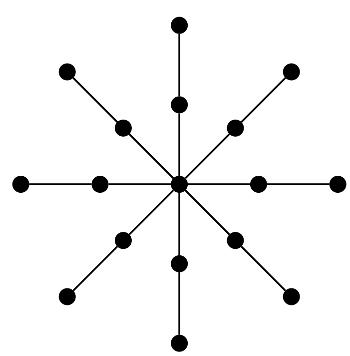
In [8], a complete classification of caterpillar trees shows that a tree converges to a metric of constant 0 curvature if and only if the tree is a caterpillar tree. It is conjectured in [8] that all trees will converge to a constant curvature metric. It then remains to find what structures of trees result in constant positive curvature or constant negative curvature. A tree has no cycles, which allows for a general expression of the Wasserstein distance, and the resulting expression for an edges curvature as defined by [8] becomes

$$k_{uv} = \frac{2 - d_u}{w_{uv} D_u} + \frac{2 - d_v}{w_{uv} D_v} \tag{4}$$

Where d_u is the degree of vertex u and $D_u := \sum_{v \sim u} \gamma(w_{uv})$ with $\gamma : \mathbb{R}_+ \to \mathbb{R}_+$ being an arbitrary one-to-one function. Furthermore, the uniqueness of any path between any pair of vertices ensures the distance condition is never violated, which is crucial for examining the evolution of the metric space under Discrete Ricci Flow. These properties were presented in [8] and used to classify convergence on caterpillar trees. In [9], a classification of a star graph results in constant zero curvature with edge weights converging to equal weights. In this paper, we consider the tree of a level 2 star graph, which results in convergence to constant negative curvature.



Star Graph (analyzed by [9])



Level 2 Star Graph

Define $G = (V, E)$ to be a level 2 star graph. Let $m = \frac{|V|-1}{2}$. A level 2 star graph is a tree with one center vertex c, m internal vertices $b_1, ..., b_m$, and m leaf vertices $a_1, ..., a_m$. An edge exists between all (b_i, a_i) pairs, and all (b_i, c) pairs. The structure is like in the image. Assume $m \geq 3$ so the graph is not a path and has at least 3 branches. We take $\gamma(x) = \frac{1}{x}$ as recommended in [8].

Proposition 1. *Under unnormalized Ollivier-Lin-Lu-Yau Ricci Flow, a level 2 star graph equipped with an initial metric converges to a limiting metric with leaf edges decaying to weight 0 and the internal edges growing to infinity, while under normalization the leaf edges decay to weight 0 and the internal edges converge to a constant limiting weight $\frac{1}{m}$ with constant negative curvature.*

Proof.

We will write $w_{b_i,c}$ to refer to any $(b_i,\ c)$ internal-center edge and w_{a_i,b_i} to refer to any (a_i, b_i) leaf-internal edge.

Using the simplified curvature expression (4) for trees we get:

$$k_{b_i,c} = \frac{2-d_c}{w_{b_i,c}D_c} + \frac{2-d_b}{w_{b_i,c}D_{b_i}} = \frac{2-m}{w_{b_i,c}D_c} + \frac{2-2}{w_{b_i,c}D_{b_i}} = \frac{2-m}{w_{b_i,c}D_c}$$

$$k_{a_i,b_i} = \frac{2-d_a}{w_{a_i,b_i}D_{a_i}} + \frac{2-d_b}{w_{a_i,b_i}D_{b_i}} = \frac{2-1}{w_{a_i,b_i}D_{a_i}} + \frac{2-2}{w_{a_i,b_i}D_{b_i}} = \frac{1}{w_{a_i,b_i}D_{a_i}} = 1$$

Where $D_c = \sum\limits_{i=1}^{m} \frac{1}{w_{b_i,c}}$ and $D_a = \frac{1}{w_{a_i,b_i}}$. The unnormalized system (1) becomes

$$\begin{cases} \frac{dw_{a_i,b_i}}{dt} = -w_{a_i,b_i} \\ \frac{dw_{b_i,c}}{dt} = \frac{m-2}{D_c} \end{cases}$$

Since the w_{ab} edges only depend on their own w_{ab} term, we can integrate directly and get

$$w_{a_i,b_i}(t) = w_{a_i,b_i}(0)e^{-t}$$

Hence the leaf edges decay to zero exponentially fast.

For the internal-center edges $(b_i,\ c)$, the curvatures do not depend on the unique $w_{b_i,c}$ edge weight, only the total sum of D_c. i.e. all internal-center edges derivatives are equal.

Let $y_{\min} := \min\limits_{1\leq j\leq m} w_{b_j,c}$ be the minimum internal-center edge weight. Then

$$D_c = \sum_{j=1}^{m} \frac{1}{w_{b_j,c}} \leq \frac{m}{y_{\min}}$$

Therefore

$$y'_{\min} = \frac{m-2}{D_c} \geq \frac{m-2}{m} y_{\min}$$

By Gronwall Inequality,

$$y_{\min}(t) \geq y_{\min}(0)e^{\frac{m-2}{m}t}$$

Since we take $m \geq 3$, then $y_{\min}$ converges to infinity as $t \to \infty$. Since we selected $y_{\min}$ as the minimal center-internal edge, it follows that all other center-internal edges $w_{b_i,c} \to \infty$ as $t \to \infty$. For any $w_{b_i,c}$ and $w_{b_j,c}$ with $i \neq j$ we have $w'_{b_i,c} - w'_{b_j,c} = \frac{m-2}{D_c} - \frac{m-2}{D_c} = 0$ hence $w_{b_i,c}(t) - w_{b_j,c}(t) = w_{b_i,c}(0) - w_{b_j,c}(0)$, i.e. their difference remains constant for all time. Let

$$C_{ij} = w_{b_i,c}(t) - w_{b_j,c}(t) = w_{b_i,c}(0) - w_{b_j,c}(0)$$

for constants C_{ij} being the associated fixed distance between two center-internal edges. Then since $w_{b_i,c}(t) = w_{b_j,c}(t) + C_{ij}$ and $w_{b_j,c}(t) \to \infty$, the ratio between two center-internal edges is

$$\frac{w_{b_i,c}(t)}{w_{b_j,c}(t)} = \frac{w_{b_j,c}(t) + C_{ij}}{w_{b_j,c}(t)} = 1 + \frac{C_{ij}}{w_{b_j,c}(t)} \to 1 \quad \text{as } t \to \infty$$

Then in the limit the curvature on internal edges is

$$k_{b_i,c} = \frac{2-m}{w_{b_i,c}D_c} = \frac{2-m}{\sum_{j=1}^{m} \frac{w_{b_i,c}}{w_{b_j,c}}} \to \frac{2-m}{m}$$

Therefore the unnormalized system results with leaf edges converging to weight 0 with curvature 1, and internal edges going to infinite weight with curvature $\frac{2-m}{m}$.

Normalization adds a term that preserves the sum of edge weights to be constant. Assume a rescaling at $t = 0$, resulting in a sum of constant 1 for all time. Our normalized system (2) becomes

$$\begin{cases} \frac{dw_{a_i,b_i}}{dt} = -w_{a_i,b_i} + w_{a_i,b_i} \sum\limits_{h \in E(G)} k_h w_h \\ \frac{dw_{b_i,c}}{dt} = \frac{m-2}{D_c} + w_{b_i,c} \sum\limits_{h \in E(G)} k_h w_h \end{cases}$$

Consider the sum of each edge type as

$$\text{leaf-internal edges} \quad X(t) := \sum_{i=1}^{m} w_{a_i,b_i} \qquad \text{internal-center edges} \quad Y(t) := \sum_{i=1}^{m} w_{b_i,c}$$

Then normalization preserves the sum $X + Y = 1$. We can write the normalization sum

$$\sum_{h \in E(G)} k_h w_h = (w_{a_1,b_1} + ... + w_{a_m,b_m}) + (\frac{2-m}{w_{b_1,c}D_c} w_{b_1,c} + ... + \frac{2-m}{w_{b_m,c}D_c} w_{b_m,c}) = X + m(\frac{2-m}{D_c})$$

Since $m \geq 3$ then $m(\frac{2-m}{D_c}) < 0$ hence we have

$$\sum_{h\in E(G)} k_h w_h < X$$

We can write

$$\frac{dw_{a_i,b_i}}{dt} = -w_{a_i,b_i} + w_{a_i,b_i} \sum_{h\in E(G)} k_h w_h = w_{a_i,b_i}([\sum k_h w_h] - 1)$$

Then we get

$$X' = \sum_{i=1}^{m} \frac{dw_{a_i,b_i}}{dt} = ([\sum k_h w_h] - 1)X = -X(1 - [\sum k_h w_h]) < -X(1-X)$$

Where in the last step the upper bound comes from our upper bound of $\sum k_h w_h < X$.

Since $X(0) > 0$ we have $X(t) > 0$ and because $X(t) + Y(t) = 1$ we also have $X(t) < 1$. Also, $X'(t) < -X(t)(1 - X(t)) < 0$, so X is decreasing and therefore converges to some limit $L \geq 0$. We will show $L = 0$ by contradiction.

Suppose $L > 0$. Then $0 < L < 1$, so for all sufficiently large time T, since $X(t)$ is decreasing we have $X(T) = L + \epsilon$ for some small positive (or possibly zero) $\epsilon \geq 0$. We can write

$$\frac{L}{2} \leq X(T) \leq \frac{1+L}{2} + \epsilon, \qquad 1 - X(T) \geq \frac{1-L}{2} + \epsilon$$

Where $\frac{L}{2} < L$ immediately gives the first lower bound. $L < 1$ gives $L < \frac{1+L}{2}$ and $X(T) = L + \epsilon < \frac{1+L}{2} + \epsilon$ gives the first upper bound. The second lower bound follows by multiplying by -1 and adding $+1$ to both sides of the first expression. Hence

$$X'(T) < -X(T)(1 - X(T)) \leq \frac{-L(1-L)}{4} - \frac{L\epsilon}{2} \leq \frac{-L(1-L)}{4} < 0$$

Integrating for some sufficiently large time T to limiting time $t >> T$,

$$X(t) - X(T) = \int_T^t X'(s)ds \leq \int_T^t \frac{-L(1-L)}{4} ds = \frac{-L(1-L)}{4}(t - T)$$

But the right-hand side goes to $-\infty$ as $t \to \infty$, contradicting $X(t) > 0$. Therefore $L = 0$ and $X(t) \to 0$. Hence under normalized flow, leaf edges converge to weight 0. Since $X(t) + Y(t) = 1$, it follows that $Y(t) \to 1$.

Denote $\hat{w}$ as unnormalized edge weight. From our unnormalized analysis, for any two internal-center edges, we have

$$\hat{w}_{b_i,c}(t) - \hat{w}_{b_j,c}(t) = \hat{w}_{b_i,c}(0) - \hat{w}_{b_j,c}(0) = C_{ij}$$

Since every internal-center unnormalized edge satisfies $\hat{w}_{b_i,c}(t) \to \infty$, the sum of all unnormalized weights also satisfies $\sum_{h \in E} \hat{w}_h(t) \to \infty$. Now compare any two normalized internal-center edges by using equation (3):

$$w_{b_i,c}(t) - w_{b_j,c}(t) = \frac{\hat{w}_{b_i,c}(t)}{\sum_{h \in E} \hat{w}_h(t)} - \frac{\hat{w}_{b_j,c}(t)}{\sum_{h \in E} \hat{w}_h(t)} = \frac{\hat{w}_{b_i,c}(t) - \hat{w}_{b_j,c}(t)}{\sum_{h \in E} \hat{w}_h(t)} = \frac{C_{ij}}{\sum_{h \in E} \hat{w}_h(t)} \to 0$$

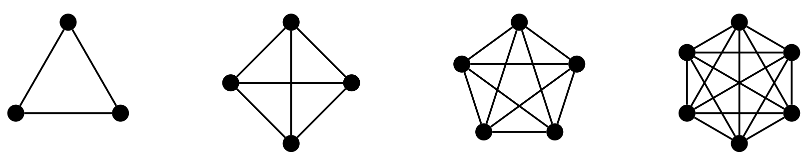
Where the last step follows from the denominator $\sum_{h \in E} \hat{w}_h(t) \to \infty$. Thus all normalized internal-center edges become asymptotically equal in the limit. Since $Y(t) = \sum_{i=1}^{m} w_{b_i,c}(t) \to 1$, and all the internal-center edge weights become equal in the limit, each one must converge to $\frac{1}{m}$. And the limiting curvature is $k_{b_i,c} = \frac{2-m}{w_{b_i,c} D_c} \to \frac{2-m}{m}$.

Therefore under unnormalized Lin-Lu-Yau flow, a level 2 star graph's leaf edges converge to weight 0 while internal edges diverge to infinite weight, and under normalized flow leaf edges converge to weight 0 while internal edges converge to weight $\frac{1}{m}$ with curvature $\frac{2-m}{m}$. For $m \geq 3$, this curvature is negative. By definition 6, we say that under normalized flow, a level 2 star graph converges to a limiting metric with constant negative curvature.

This completes the proof of **Proposition 1**.

Complete Graph

In the next sections, we consider graphs with cycles. The difficulty of analyzing cyclic graphs compared to trees is due to there being no known simplification for the Wasserstein distance, whereas we have equation (4) for trees. The resulting ODE systems are often more complex as well. We also need to ensure the distance condition is never violated, which is another challenge that emerges in cyclic graph analysis. We present 3 different types of cyclic graphs, 2 of which the results completely classify the limiting behavior (Propositions 2 and 3), and 1 of which results give a partial classification of the limiting behavior (Proposition 4).



K_3 K_4 K_5 K_6

A trivial graph structure for Discrete Ricci Flow is a complete graph with an initial metric of all edge weights being equal. Intuitively, the fully symmetrical structure results in a perfectly even distribution of mass in the transportation problem, resulting in equal positive curvatures across all edges. Then under unnormalized flow, the edge weights collapse to weight 0 while normalization preserves the initial metric. The following analysis of this trivial case can demonstrate how analyzing convergence on general graphs can be done for other non-trivial cases.

Proposition 2. *For a complete graph $K_n = (V, E)$ equipped with an equal-weighted initial metric, under unnormalized Lin-Lu-Yau Flow the edge weights converge to a metric with all 0 edge weights, while normalization preserves the initial metric with constant positive curvature $\frac{n}{n-1}$.*

Proof.

A complete graph $K_n = (V, E)$ is a graph with n vertices, and every pair of vertices (u, v) has an edge between them, hence we have $|E| = \binom{n}{2} = \frac{n(n-1)}{2}$ edges. Under an initially equal metric, define $w_i(0) := w_1(0) = w_2(0) = ... = w_{|E|}(0)$. Since there is a direct edge between all pairs of nodes, $d_{u,v}(0) = w_{u,v}(0)$ for all edges, and our distance condition will never be violated. Hence $d_i(0) = d_1(0) = d_2(0) = ... = d_{|E|}(0) = w_i(0)$. We will show that $w_i(t)$ goes to 0 under unnormalized flow, while under normalized flow $w_i(t) = w_i(0)$ for all time, i.e. $w_i(t)$ does not change from the initial metric.

Since an edge exists between all pairs of nodes, the probability distribution (Definition 1) for any fixed node u over all nodes $x \in V$ at time $t = 0$ simplifies to

$$\mu_u^\alpha(x) = \alpha \quad \text{if } x = u \qquad \mu_u^\alpha(x) = (1-\alpha)\frac{\gamma(w_i(0))}{(n-1)\gamma(w_i(0))} = \frac{1-\alpha}{n-1} \quad \text{otherwise}$$

Since LLY curvature takes the limit as $\alpha \to 1$, Assume $\alpha > \frac{1}{n}$. Fix two nodes u and v. Then the Wasserstein distance associated with edge (u, v) is $W(\mu_u, \mu_v)$. Over all pairs of vertices $x, y \in V$ Consider the coupling (Definition 2) as

$$A(x,y) = \begin{cases} \frac{1-\alpha}{n-1} & \text{if } x = y \\ \alpha - \frac{1-\alpha}{n-1} & \text{if } x = u, \ y = v \\ 0 & \text{otherwise} \end{cases}$$

Where the coupling determines how much mass to send from x to y, i.e. a transportation plan for the transportation problem. Now,

$$\sum_{y \in V} A(x,y) = \alpha - \frac{1-\alpha}{n-1} + \frac{1-\alpha}{n-1} = \alpha = \mu_u(x) \quad \text{if } x = u$$

$$\sum_{y \in V} A(x,y) = \frac{1-\alpha}{n-1} = \mu_u(x) \quad \text{if } x \neq u$$

$$\sum_{x \in V} A(x,y) = \alpha - \frac{1-\alpha}{n-1} + \frac{1-\alpha}{n-1} = \alpha = \mu_v(y) \quad \text{if } y = v$$

$$\sum_{x \in V} A(x,y) = \frac{1-\alpha}{n-1} = \mu_v(y) \quad \text{if } y \neq v$$

Which shows this coupling is feasible.

To ensure optimality, we consider the dual problem (Definition 3). Consider the function

$$g(x) = w_i(0) \quad \text{if } x = u, \qquad\qquad g(x) = 0 \quad \text{otherwise.}$$

$$\text{When } x = u \text{ or } y = u \text{ and } x \neq y, \quad |g(x) - g(y)| = w_i(0) = d_i(0)$$

$$\text{Otherwise,} \quad |g(x) - g(y)| = 0 \leq d_i(0)$$

Therefore $g(x)$ is 1-Lipschitz, which is feasible for the dual problem.

Comparing the costs of the feasible coupling $A(x, y)$ with the feasible dual problem given by $g(x)$,

$$W(\mu_u, \mu_v) = \sum_{x,y \in V} A(x, y)d(x, y) = w_i(0)\frac{n\alpha - 1}{n - 1}$$

$$W(\mu_u, \mu_v) = \sum_{x \in V} g(x)[\mu_u(x) - \mu_v(x)] = w_i(0)\frac{n\alpha - 1}{n - 1}$$

Since both result in the same cost, by duality we have an optimal solution.

Our α-curvature (Definition 4) at $t = 0$ becomes,

$$k_\alpha(u, v) = 1 - \frac{W(\mu_u^\alpha, \mu_v^\alpha)}{d_{u,v}} = 1 - \frac{w_i(0)\frac{n\alpha-1}{n-1}}{w_i(0)} = \frac{n(1 - \alpha)}{n - 1}$$

The LLY curvature at time $t = 0$ is

$$k(u, v) = \lim_{\alpha \to 1} \frac{k_\alpha(u, v)}{1 - \alpha} = \frac{n}{n - 1}$$

This shows that at $t = 0$ all edges have the same curvature. Since all edges are initially equal, all pairs of nodes are connected, and the curvatures are initially equal, the system is fully symmetrical and will evolve symmetrically. By the uniqueness and existence of solutions to ODEs, the above curvature computation will apply to all edges for all time. Therefore if $w_i(0) = w_j(0)$ for all edges i, j, then $w_i(t) = w_j(t)$ for all $t \geq 0$. Hence, starting from an equal initial metric, all edge curvatures remain equal for all time.

Our unnormalized flow is given by

$$\frac{dw_{xy}(t)}{dt} = -k_{xy}(t)w_{xy}(t)$$

Where a solution is

$$w_{xy}(t) = w_{xy}(0)\exp(\int -k_{xy}dt) = w_{xy}(0)\exp(\frac{-nt}{n - 1})$$

With $n \geq 3$, this results in exponential decay to 0.

Our normalized flow is given by

$$\frac{dw_{xy}(t)}{dt} = -k_{xy}(t)w_{xy}(t) + w_{xy}(t) \sum_{e \in E(G)} k_e w_e(t)$$

Where $\sum_{e \in E(G)} k_e w_e(t) = \frac{n}{n-1}$ since all curvatures are the same and the sum of edge weights remains constant 1 by normalization, assuming a rescaling at time $t = 0$ such that the initial metric sums to constant 1. Hence,

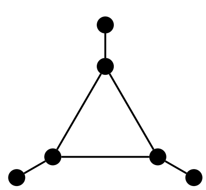
$$\frac{dw_{xy}(t)}{dt} = -k_{xy}(t)w_{xy}(t) + w_{xy}(t) \sum_{e \in E(G)} k_e w_e(t) = -\frac{n}{n-1} w_{xy}(t) + w_{xy}(t) \frac{n}{n-1} = 0$$

So, the derivative is 0 for all time, meaning the edge weights do not change from the initial metric.

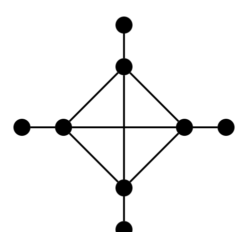
In conclusion, the Lin-Lu-Yau Ricci Curvature on Complete Graphs with n vertices converges to constant positive curvature of $\frac{n}{n-1}$. Under unnormalized flow, the metric collapses to all edge weights being 0, while normalization results in the metric remaining unchanged for all time, i.e. $w_i(0) = w_i(t)$ for all edges for all time. This completes the proof of **Proposition 2**.

Complete Graph With Additional Edges

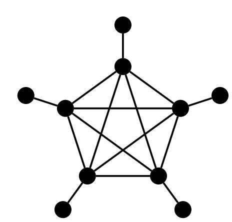
Next we consider a graph formed by a complete K_n graph and appending one edge for each node. Similarly to the complete graph analysis, under an initially equal metric, symmetry is induced which creates a simplified structure.



K_3 with 3 pendant edges



K_4 with 4 pendant edges



K_5 with 5 pendant edges

Proposition 3. *Let K_n^* be a graph resulting from modifying a complete $K_n = (V, E)$ graph by appending n edges and n nodes, one for each original node. Equipped with an initially equal metric, unnormalized flow results in all edges collapsing to 0 weights. Under normalized flow, the graph converges to constant curvature, with curvature value and edge weight values depending on the size of n.*

Proof.

Initial Metric

A complete K_n graph has $\frac{n(n-1)}{2}$ edges, hence our modified K_n^* graph has $\frac{n(n-1)}{2} + n$ edges. Note that n refers to the number of nodes in the original K_n graph. Let the initial metric be all edge weights initially equal,

$$w_{ij}(0) = w_{xy}(0) \quad \forall (i,j),(x,y) \in E$$

Symmetry between edges

Since the initial metric has all equal edge weights, the symmetrical structure of the graph induces symmetry between internal edges and leaf edges. Namely, we can classify two types of edges by the degrees of the nodes they connect. Let w_i refer to any internal edge connecting two degree n nodes, and w_l refer to any leaf edge connecting a degree n node with a degree 1 node. This classification defines every edge in the graph. The initially equal metric will result in an initially equal curvature within the two edge classes. Hence by the uniqueness and existence of solutions to ODEs, any two $w_i(t)$ edges will remain equal and any two $w_l(t)$ edges will remain equal.

Distance Metric

If two nodes x and y where $(x, y) \in E$ have a shortest distance that is not the edge weight w_{xy} at any point in time, then a distance violation is triggered. For leaf-internal edges the distance violation will never occur since any path must include the edge weight itself. For internal edges, a distance violation cannot occur since under the symmetrical metric, at $t = 0$ it is not violated and the internal edge weights will evolve symmetrically preserving this fact.

Hence for all time, the shortest distance between any two nodes that have an edge between them is the edge weight, i.e. our distance metric is never violated. We use $d(x)$ to refer to the degree of a node, while $d(x, y)$ refers to the distance metric between two edges. From this it follows that

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ w_i & \text{if } d(x) = n, \quad d(y) = n \\ w_l & \text{if } d(x) = n, \quad d(y) = 1, \quad x \sim y \\ w_l + w_i & \text{if } d(x) = n, \quad d(y) = 1, \quad x \nsim y \\ 2w_l + w_i & \text{if } d(x) = 1, \quad d(y) = 1 \end{cases}$$

The mass distributions (Definition 1.) are constructed as follows.

Consider any internal node u where $d(u) = n$. u has exactly 1 adjacent leaf node and exactly $n-1$ adjacent internal nodes. Then, define

$$T_i := \mu_u(x) = \frac{(1-\alpha)\gamma(w_i)}{(n-1)\gamma(w_i) + \gamma(w_l)} \quad \text{if } d(x) = n \text{ and } x \sim u$$

$$T_l := \mu_u(x) = \frac{(1-\alpha)\gamma(w_l)}{(n-1)\gamma(w_i) + \gamma(w_l)} \quad \text{if } d(x) = 1 \text{ and } x \sim u$$

Consider any leaf node v where $d(v) = 1$. v has exactly 1 adjacent internal node. Then

$$\mu_v(x) = (1-\alpha) \quad \text{if } x \sim u$$

And we have that $\mu_u(u) = \mu_v(v) = \alpha$.

Using the above simplifications, fix a,b as any two adjacent degree n nodes, and c as the degree 1 node adjacent to a, and d as the degree 1 node adjacent to b. Then the mass distributions are as follows.

$$\mu_a(a) = \alpha \quad \mu_a(b) = T_i \quad \mu_a(c) = T_l \quad \mu_a(d) = 0 \quad \mu_a(x) = T_i \text{ if } x \sim a, x \neq c \quad\quad \mu_a(x) = 0 \quad \text{otherwise}$$

$$\mu_b(a) = T_i \quad \mu_b(b) = \alpha \quad \mu_b(c) = 0 \quad \mu_b(d) = T_l \quad \mu_b(x) = T_i \text{ if } x \sim b, x \neq d \quad\quad \mu_b(x) = 0 \quad \text{otherwise}$$

$$\mu_c(a) = (1-\alpha) \quad \mu_c(b) = 0 \quad \mu_c(c) = \alpha \quad \mu_c(d) = 0 \quad \mu_c(x) = 0 \quad \text{otherwise}$$

$$\mu_d(a) = 0 \quad \mu_d(b) = (1-\alpha) \quad \mu_d(c) = 0 \quad \mu_d(d) = \alpha \quad \mu_d(x) = 0 \quad \text{otherwise}$$

Due to symmetry, we only need to consider $W(\mu_a, \mu_b)$ and $W(\mu_a, \mu_c)$ which represent the transportation cost for our two edge types w_i and w_l respectively.

For $W(\mu_a, \mu_b)$ take the transportation plan A and Kantorovich potential function $g(x)$ as

$$A(x,y) = \begin{cases} T_i & \text{if } x = y,\ d(x) = d(y) = n \\ \alpha - T_i & \text{if } (x,y) = (a,b) \\ T_l & \text{if } (x,y) = (c,d) \\ 0 & \text{otherwise} \end{cases} \qquad g(x) = \begin{cases} w_i & \text{if } x = a \\ w_i + w_l & \text{if } x = c \\ -w_l & \text{if } x = d \\ 0 & \text{otherwise} \end{cases}$$

It is straightforward to verify A and $g(x)$ are feasible and both yield the same transportation cost:

$$W(\mu_a, \mu_b) = (\alpha - T_i)w_i + (T_l)(w_i + 2w_l)$$

Which results in the k-alpha curvature

$$k_{ab}^{\alpha} = 1 - \frac{W(\mu_a, \mu_b)}{d(a,b)} = 1 - \frac{(\alpha - T_i)w_i + (T_l)(w_i + 2w_l)}{w_i} = 1 - \alpha + T_i - T_l - \frac{2w_l}{w_i}T_l$$

Defining $k_{w_i} := k_{ab}$ which is equal to any internal edge, and expanding T_i and T_l terms we get

$$k_{w_i} := k_{ab} = \lim_{\alpha \to 1} \frac{k_{ab}^{\alpha}}{1-\alpha} = \frac{nw_i\gamma(w_i) - 2w_l\gamma(w_l)}{w_i((n-1)\gamma(w_i) + \gamma(w_l))}$$

For $W(\mu_a, \mu_c)$ take the transportation plan A and Kantorovich potential function $g(x)$ as

$$A(x,y)=\begin{cases}(1-\alpha) & \text{if } (x,y)=(a,a)\\ (2\alpha-1) & \text{if } (x,y)=(a,c)\\ T_i & \text{if } x\neq a,\quad y=c,\quad d(x)=n\\ T_l & \text{if } (x,y)=(c,c)\\ 0 & \text{otherwise}\end{cases}\qquad g(x)=\begin{cases}w_l & \text{if } x=a\\ 0 & \text{if } x=c\\ w_i+w_l & \text{otherwise}\end{cases}$$

It is straightforward to verify A and $g(x)$ are feasible and both yield the same transportation cost:

$$W(\mu_a,\mu_c)=(2\alpha-1)w_l+(n-1)T_i(w_l+w_i)$$

Which results in the k-alpha curvature

$$k^{\alpha}_{ac}=1-\frac{W(\mu_a,\mu_c)}{d(a,c)}=1-\frac{(2\alpha-1)w_l+(n-1)T_i(w_l+w_i)}{w_l}=2(1-\alpha)-(n-1)T_i(1+\frac{w_i}{w_l})$$

Defining $k_{w_l}:=k_{ac}$ which is equal to any leaf edge, and expanding T_i we get

$$k_{w_l}:=k_{ac}=\lim_{\alpha\to 1}\frac{k^{\alpha}_{ac}}{1-\alpha}=\frac{(n-1)w_l\gamma(w_i)+2w_l\gamma(w_l)-(n-1)w_i\gamma(w_i)}{w_l((n-1)\gamma(w_i)+\gamma(w_l))}$$

To analyze the long time behavior of our system, we need to choose a gamma function. We will choose $\gamma(x)=\frac{1}{x}$ as our gamma function as suggested by [8]. The unnormalized system (1) becomes

$$\begin{cases}\frac{dw_l}{dt}=-\frac{(n-1)w_l-(n-3)w_i}{(n-1)w_l+w_i}(w_l)\\ \frac{dw_i}{dt}=-\frac{(n-2)w_l}{(n-1)w_l+w_i}(w_i)\end{cases}$$

We perform an analysis of the long time behavior of the unnormalized system above as follows. Recall that the graph is defined for $n\geq 3$. Consider the ratio $r(t):=\frac{w_i(t)}{w_l(t)}$. Then

$$\frac{dr}{dt}=\frac{w_i'w_l-w_iw_l'}{w_l^2}=\frac{-k_iw_iw_l+k_lw_iw_l}{w_l^2}=\frac{w_i}{w_l}(k_l-k_i)=r(k_l-k_i)=r(\frac{w_l-(n-3)w_i}{(n-1)w_l+w_i})=\frac{r(1-(n-3)r)}{n-1+r}$$

Where we write w_i as $w_i=rw_l$ to obtain the final simplified form above. Since our initial metric is all equal edge weights, we have $r(0)=\frac{w_i(0)}{w_l(0)}=1$. For the case when $n=4$, we have $r'=\frac{r(1-r)}{3+r}$. If $r(t)=1$ for all time t, then $r'(t)=0$ and $\frac{r(t)(1-r(t))}{3+r(t)}=0$. Therefore $r(t)=1$ for all time is a solution. Then by uniqueness and existence of solutions to ODEs, then $r(t)=1$. Hence $w_i(t)=w_l(t)$, allowing to solve directly for $w_l'=-\frac{1}{2}w_l$. Then $w_l(t)=w_i(t)=w_l(0)e^{-t/2}$.

It remains to analyze the cases when $n=3$ and $n\geq 5$. Since for these cases $r>0$ and

$1-(n-3)r \neq 0$, we can separate variables in $\frac{dr}{dt}$ and integrate

$$\frac{dr}{dt} = \frac{r(1-(n-3)r)}{n-1+r} \quad \Longrightarrow \quad \frac{n-1+r}{r(1-(n-3)r)}dr = dt \quad \Longrightarrow \quad \int \frac{n-1+r}{r(1-(n-3)r)}dr = \int 1 \; dt$$

For $n=3$, integrating we get

$$2\ln(r) + r = t + C$$

we can use $r(0) = 1$ to solve for $C = 1$. Then

$$2\ln(r) + r = t + 1$$

We have $r'(t) = \frac{r}{2+r} > 0$, hence $r(t)$ is increasing. As $t \to \infty$, then $t+1 \to \infty$, forcing $r(t) \to \infty$.

For $n \geq 5$, integrating we get

$$(n-1)\ln(r) - \frac{(n-2)^2}{n-3}\ln|1-(n-3)r| = t + C$$

Solving $r' = 0$ gives the positive equilibrium $r = \frac{1}{n-3}$. Since $r(0) = 1$ and $\frac{1}{n-3} < 1$ for $n \geq 5$, the solution always starts above this equilibrium. Whenever $r(t) > \frac{1}{n-3}$, we have $1-(n-3)r < 0$, so $r'(t) < 0$. Hence $r(t)$ is decreasing and by uniqueness of solutions, $r(t)$ cannot cross the equilibrium. Therefore $\frac{1}{n-3} < r(t) < 1$. Then the $(n-1)\ln(r)$ term is bounded, and we also have $\frac{(n-2)^2}{n-3} > 0$. As $t \to \infty$ we have $t + C \to \infty$. Therefore it must be the case that $\ln|1-(n-3)r| \to -\infty$, thus $(1-(n-3r)) \to 0$. Therefore $r \to \frac{1}{n-3}$.

We have classified the behavior of $r(t)$ for both cases of $n = 3$ and $n \geq 5$. We will use the limit of $r(t)$ to show the behavior of the edge weights. We can express $\frac{dw_l}{dr}$ as the following

$$\frac{dw_l}{dt} = \frac{dw_l}{dr}\frac{dr}{dt} \quad \Longrightarrow \quad \frac{dw_l}{dr} = \frac{w_l'}{r'} = -\frac{((n-1)-(n-3)r)w_l}{r(1-(n-3)r)} \quad \Longrightarrow \quad \frac{1}{w_l}dw_l = -\frac{(n-1)-(n-3)r}{r(1-(n-3)r)}dr$$

Where the last step comes from substituting our expressions for w_l' and r' and $w_i = rw_l$. Also $r'(t) > 0$ if $n = 3$ and $r'(t) < 0$ if $n \geq 5$, so we are not dividing by 0.

$$\frac{1}{w_l}dw_l = -\frac{(n-1)-(n-3)r}{r(1-(n-3)r)}dr \quad \Longrightarrow \quad \begin{cases} \frac{1}{w_l}dw_l = -\frac{2}{r}dr & \text{if } n = 3 \\ \ln w_l = -(n-1)\ln r + (n-2)\ln|1-(n-3)r| + C & \text{if } n \geq 5 \end{cases}$$

For $n = 3$, integrating we get

$$\int \frac{1}{w_l}dw_l = \int -2\frac{1}{r}dr \quad \Longrightarrow \quad \ln w_l = -2\ln r + C \quad \Longrightarrow \quad w_l = Cr^{-2} \quad \text{if } n = 3$$

In the last step we exponentiate both sides and write e^C as C since it is some constant.

And since $r(0) = 1$ we get $C = w_l(0)$

Hence

$$w_l(t) = \frac{w_l(0)}{r(t)^2} \quad , \quad w_i(t) = r(t)w_l(t) = \frac{w_i(0)}{r(t)} = \frac{w_l(0)}{r(t)}$$

Since $r(t) \to \infty$, both edge weights decay to weight 0.

For $n \geq 5$, exponentiating we get,

$$w_l(t) = Cr(t)^{-(n-1)}|1 - (n-3)r(t)|^{n-2}$$

Using $r(0) = 1$ and $n \geq 5$,

$$C = \frac{w_l(0)}{(n-4)^{n-2}}$$

Hence

$$w_l(t) = \frac{w_l(0)r(t)^{-(n-1)}}{(n-4)^{n-2}}|1 - (n-3)r(t)|^{n-2}$$

Since $r(t) \to \frac{1}{n-3}$, we have $r(t)^{-(n-1)}$ approaches a positive finite number, but $|1-(n-3)r(t)|^{n-2}$ approaches 0. Therefore $w_l(t) \to 0$ and $w_i(t) = r(t)w_l(t) \to 0$. Therefore, under unnormalized flow, for any $n \geq 3$ all edge weights decay to 0 with the ratio $r(t) := \frac{w_i(t)}{w_l(t)}$ dictating the rate of decay of w_i compared to w_l by $w_i(t) = r(t)w_l(t)$.

$$\begin{cases} w_l(t) = \frac{w_l(0)}{r(t)^2} \quad r(t) \to \infty & \text{if } n = 3 \\ w_l(t) = w_l(0)e^{-t/2} \quad r(t) \to 1 & \text{if } n = 4 \\ w_l(t) = \frac{w_l(0)r(t)^{-(n-1)}}{(n-4)^{n-2}}|1-(n-3)r(t)|^{n-2} \quad r(t) \to \frac{1}{n-3} & \text{if } n \geq 5 \end{cases}$$

Introducing normalization, the system becomes

$$\begin{cases} \frac{dw_l}{dt} &= -\frac{(n-1)w_l-(n-3)w_i}{(n-1)w_l+w_i}(w_l) + w_l \sum\limits_h k_h w_h \\ \frac{dw_i}{dt} &= -\frac{(n-2)w_l}{(n-1)w_l+w_i}(w_i) + w_i \sum\limits_h k_h w_h \end{cases}$$

Consider the ratio $r = \frac{w_i}{w_l}$. Then

$$\frac{dr}{dt} = \frac{w_i'w_l - w_iw_l'}{w_l^2} = \frac{(\hat{w}_i' + \hat{w}_i \sum\limits_h k_h\hat{w}_h)\hat{w}_l - (\hat{w}_l' + \hat{w}_l \sum\limits_h k_h\hat{w}_h)\hat{w}_i}{\hat{w}_l^2} = \frac{\hat{w}_i'\hat{w}_l - \hat{w}_l'\hat{w}_i}{\hat{w}_l^2} = \frac{r(1-(n-3)r)}{n-1+r}$$

Where $\hat{w}_i$, $\hat{w}_l$, and $\hat{w}_h$ denote the unnormalized version of our edge weights, and we express the normalized version by writing the unnormalized weights directly. Since the normalization terms cancel, we result with the same expression for $\frac{dr}{dt}$ in the normalized system as in the unnormalized system. In the graph K_n^*, we have $\frac{n(n-1)}{2}$ internal edges and n leaf edges. Let M be the number of internal edges. Then assuming a rescaling at $t = 0$, normalization preserves the sum $nw_l + Mw_i = 1$.

Then since $w_i = rw_l$, we get

$$w_l(t) = \frac{1}{n + Mr(t)} \qquad w_i(t) = \frac{r(t)}{n + Mr(t)}$$

Where substituting $w_i = rw_l$ into our curvature expressions we get

$$k_{w_l}(t) = \frac{(n-1) - (n-3)r}{n - 1 + r} \qquad k_{w_i}(t) = \frac{n-2}{n-1+r}$$

So the normalized convergence follows directly from the limit of $r(t)$.

For $n = 3$, we have $r(t) \to \infty$ and $M = 3$. Then

$$w_l(t) = \frac{1}{3 + 3r(t)} \to 0 \qquad w_i(t) = \frac{r(t)}{3(1 + r(t))} \to \frac{1}{3} \qquad k_{w_i}(t) = k_{w_l}(t) \to 0$$

For $n = 4$, we have $r(t) \to 1$ and $M = 6$. Then

$$w_l(t) = \frac{1}{4 + 6r(t)} \to \frac{1}{10} \qquad w_i(t) = \frac{r(t)}{4 + 6r(t)} \to \frac{1}{10} \qquad k_{w_i}(t) = k_{w_l}(t) \to \frac{1}{2}$$

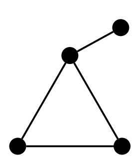
For $n \geq 5$, we have $r(t) \to \frac{1}{n-3}$ and $M = \frac{n(n-1)}{2}$. Then

$$w_l(t) \to \frac{2(n-3)}{n(3n-7)} \qquad w_i(t) \to \frac{2}{n(3n-7)} \qquad k_{w_i}(t) = k_{w_l}(t) \to \frac{n-3}{n-2}$$

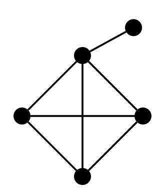
Therefore under unnormalized flow, any K_n^* graph results in all edges decaying to weight 0, while under normalization the edge weights converge to a positive limiting metric with constant positive curvature. The value of the curvature and the resulting limiting metric edge weights depend on the size n of the graph, as described above. This completes the proof of **Proposition 3**.

Complete Graph Plus One More Edge

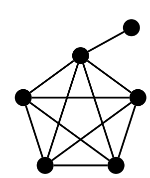
Next we consider a different modification to a complete K_n graph by appending one additional edge and node.



K_3 with 1 pendant edge



K_4 with 1 pendant edge



K_5 with 1 pendant edge

Let K_n^+ be a complete K_n graph with one more vertex and one more edge connecting the additional vertex to exactly one other vertex in the graph. The structure is like the image above. We will perform an analysis on the long time behavior of the edge weights in K_n^+ under Lin-Lu-Yau

Ricci flow. Our results give a partial classification of the limiting metric, identifying one edge types limiting behavior under unnormalized flow.

Proposition 4. *A K_n^+ graph equipped with an initial metric of all edge weights being equal results in any edges connecting two degree $n-1$ nodes collapsing to weight 0 under unnormalized Lin-Lu-Yau flow.*

Proof.

Initial Metric

A complete K_n graph has $\frac{n(n-1)}{2}$ edges, hence our modified K_n^+ graph has $\frac{n(n-1)}{2}+1$ edges. Let the initial metric be all edge weights initially equal,

$$w_{ij}(0) = w_{xy}(0) \quad \forall (i,j),(x,y) \in E$$

Symmetry between edges

Since the initial metric has all equal edge weights, the structure of the graph induces symmetry. We have exactly 1 degree n node, let this node be c. We have exactly 1 degree 1 node, let this node be d. Then let the $n-2 =: m$ remaining nodes of degree $n-1$ be a_i for $i = 1, ...m$. Let w_x refer to any edge connecting two degree $n-1$ nodes a_i, a_j. Let w_y refer to any edge connecting a degree $n-1$ node a_i with the degree n node c. Let w_z refer to the one edge connecting the degree n node c with the degree 1 node d. This classification defines every edge in the graph. The initially equal metric will result in an initially equal curvature within the three edge classes. Hence by the symmetry, any two edges in the same class will remain equal weights for all time t.

Distance Metric

Since a path between nodes d,c always includes w_z, we have $d(d,c) = w_z$ for all time. Since a path between nodes a_i, c always includes a w_y edge, we have $d(a_i,c) = w_{a_i,c} = w_y$ for all time. For fixed nodes a_i, a_j, a path with $w_{a_u,a_v} = w_x$ with $u, v \neq i, j$ must include at least one more w_x or w_y edge, hence w_{a_i,a_j} will be shorter. It remains to consider the case for fixed nodes a_i, a_j with a path including w_y. We have any $w_y + w_x > w_x$. Then the only possible distance condition violation is if a path $2w_y < w_x$ exists between nodes a_i, a_j.

If at any point in time we have $2w_y < w_x$ such that the shortest distance $d(a_i, a_j) = 2w_y \neq w_x$, then a distance condition violation is triggered. In this case, all edges of type w_x must be deleted to proceed with Ricci Flow evolution. Thus we remain with a star graph centered at vertex c, and [9]) shows convergence results for any star graph.

It then remains to consider the case for which all time, $w_{uv} = d(u,v)$, meaning the shortest distance between two nodes is the edge weight if an edge exists, i.e we never violate the distance condition. For our graph, this means $w_x \leq 2w_y$ for all time. So, we continue assuming $w_x \leq 2w_y$ for all time.

Fix one edge (a_i, a_j) and let $a = a_i$ and $b = a_j$. Then let any a_k refer to a node of type a for $k \neq i \neq j$. For our four fixed nodes a, b, c, d the mass distributions are as follows.

Let

$$T_1 := \frac{(1-\alpha)}{(n-2)\gamma(w_x) + \gamma(w_y)} \qquad T_2 := \frac{(1-\alpha)}{(n-1)\gamma(w_y) + \gamma(w_z)}$$

Then we have

$$\mu_a(a) = \alpha \quad \mu_a(b) = T_1\gamma(w_x) \quad \mu_a(a_k) = T_1\gamma(w_x) \quad \mu_a(c) = T_1\gamma(w_y) \quad \mu_a(d) = 0$$

$$\mu_b(a) = T_1\gamma(w_x) \quad \mu_b(b) = \alpha \quad \mu_b(a_k) = T_1\gamma(w_x) \quad \mu_b(c) = T_1\gamma(w_y) \quad \mu_b(d) = 0$$

$$\mu_c(a) = T_2\gamma(w_y) \quad \mu_c(b) = T_2\gamma(w_y) \quad \mu_c(a_k) = T_2\gamma(w_y) \quad \mu_c(c) = \alpha \quad \mu_c(d) = T_2\gamma(w_z)$$

$$\mu_d(a) = 0 \quad \mu_d(b) = 0 \quad \mu_d(a_k) = 0 \quad \mu_d(c) = (1-\alpha) \quad \mu_d(d) = \alpha$$

Using the assumption that $w_{uv} = d_{uv}$ for all time if an edge exists between nodes (u, v), the distance metric for all $d(u, v)$ pairs is given by:

$$d(u,v) = \begin{cases} w_{uv} & \text{if } (u,v) \in E \\ w_y + w_z & \text{if } u = a_k, \quad v = d \\ 0 & \text{if } u = v \end{cases}$$

To define the curvature on each edge type, we need the Wasserstein distance on each edge type w_x, w_y, w_z. For edge w_z we consider $W(\mu_c, \mu_d)$. An optimal transportation plan is given by

$$A(x,y) = \begin{cases} T_2\gamma(w_y) & \text{if } x = a, b, a_k, \quad y = c \\ T_2\gamma(w_z) & \text{if } (x,y) = (c,c), (d,d) \\ \alpha - T_2\gamma(w_z) & \text{if } (x,y) = (c,d) \\ 0 & \text{otherwise} \end{cases} \qquad g(x) = \begin{cases} w_y & \text{if } x = a, b, a_k \\ -w_z & \text{if } x = d \\ 0 & \text{otherwise} \end{cases}$$

It is straightforward to verify $A(x, y)$ and $g(x)$ are feasible and result in the cost:

$$W(\mu_c, \mu_d) = (n-1)T_2\gamma(w_y)w_y + (\alpha - T_2\gamma(w_z))w_z$$

Then our alpha curvature is

$$k^{\alpha}_{cd} = 1 - \frac{W(\mu_c, \mu_d)}{d(c,d)} = 1 - \alpha + T_2\gamma(w_z) - (n-1)T_2\gamma(w_y)\frac{w_y}{w_z}$$

Expanding T_2, the Lin-Lu-Yau curvature is

$$k_{cd} = 1 + \frac{\gamma(w_z)}{(n-1)\gamma(w_y) + \gamma(w_z)} - \frac{(n-1)w_y\gamma(w_y)}{w_z((n-1)\gamma(w_y) + \gamma(w_z))}$$

For edge w_x we consider $W(\mu_a, \mu_b)$. An optimal transportation plan is given by

$$A(x,y) = \begin{cases} T_1\gamma(w_x) & \text{if } (x,y) = (a,a), (b,b), (a_k, a_k) \\ \alpha - T_1\gamma(w_x) & \text{if } (x,y) = (a,b) \\ T_1\gamma(w_y) & \text{if } (x,y) = (c,c) \\ 0 & \text{otherwise} \end{cases} \qquad g(x) = \begin{cases} w_x & \text{if } x = a \\ 0 & \text{if } x = b \\ \frac{w_x}{2} & \text{otherwise} \end{cases}$$

It is straightforward to verify $A(x,y)$ and $g(x)$ are feasible and result in the cost:

$$W(\mu_a, \mu_b) = (\alpha - T_1\gamma(w_x))w_x$$

Then

$$k_{ab}^{\alpha} = 1 - \frac{W(\mu_a, \mu_b)}{d(a,b)} = 1 - \alpha + T_1\gamma(w_x)$$

And expanding T_1 and writing $k_x := k_{ab}$, we get

$$k_x := k_{ab} = \lim_{\alpha \to 1} \frac{k_{ab}^{\alpha}}{1-\alpha} = 1 + \frac{\gamma(w_x)}{(n-2)\gamma(w_x) + \gamma(w_y)}$$

Intermediate results

Without considering k_z, we can deduce the following from k_x alone. Since $\gamma : \mathbb{R}_+ \to \mathbb{R}_+$ and all edge weights remain positive for all finite time, we have

$$0 < \frac{\gamma(w_x)}{(n-2)\gamma(w_x) + \gamma(w_y)} < \frac{1}{n-2}$$

Hence

$$1 < k_x < 1 + \frac{1}{n-2}$$

Which gives a bound for the solution of unnormalized flow $w_x(t) = -k_x w_x$ as

$$-(1 + \frac{1}{n-2})w_x < w_x' < -w_x$$

Integrating gives

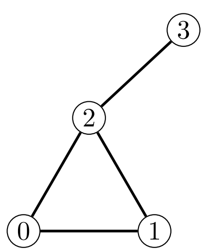
$$w_x(0)\exp(-(1 + \frac{1}{n-2})t) < w_x(t) < w_x(0)\exp(-t)$$

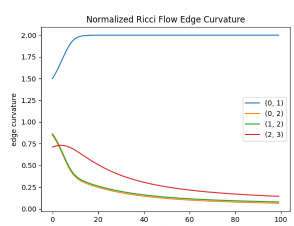
Therefore for $n \geq 3$ we have $w_{ab}(t) \to 0$ exponentially fast as $t \to \infty$ under unnormalized Lin-Lu-Yau curvature. This result is independent of the choice for α and $\gamma(x)$. This completes the proof for **Proposition 4**, and gives a partial classification of the limiting behavior of a K_n^+ graph. It remains to define the curvature of k_y edges and classify the behavior of the w_y edges and w_z edges.

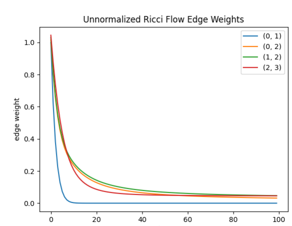
Experimental Results

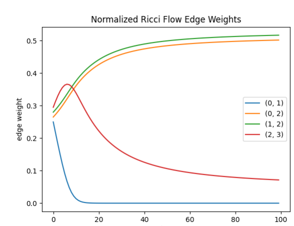
We present experimental results of normalized and unnormalized Ricci flow on three graphs: G_1, a complete 3 graph with one pendant edge. G_2, a complete 4 graph with one pendant edge. G_3, a complete 3 graph with two pendant edges. The results were produced by numerically approximating the Ricci flow. At each discrete time step, we computed the Lin-Lu-Yau curvature of each edge by solving the associated optimal transportation problem. The edge weights were then updated using an Euler step with step size $\delta = 0.25$. See supplementary information in [9] for the detailed algorithm description as designed by the authors. We ran 100 iterations for each graph, corresponding to simulating the flow up to time $t = 25$. The exact implementation we used can be found at https://github.com/andrewely8/Discrete-Ricci-Flow-Research-Project .

All experiments are done with initially equal edge weights. A slight shift to each line is applied in the visual plots to prevent overlap.

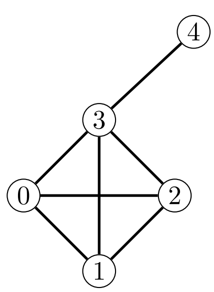


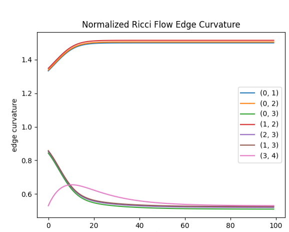


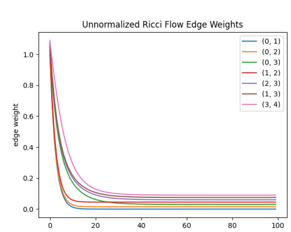


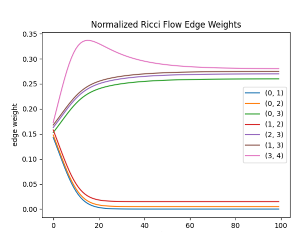


G_1



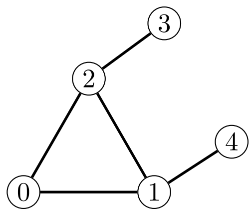




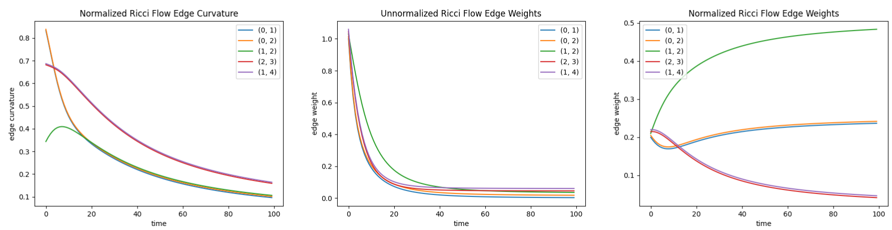


G_2

The results of G_1 and G_2 show the edges connecting any two degree n nodes decay to weight 0, as expected by the partial results in **Proposition 4**. In both G_1 and G_2, the unnormalized edge weights all decay to weight 0. under normalized flow, G_1's pendant edge and the edge connecting two degree n nodes decay to weight 0 while the other two edges converge to a positive limiting weight with constant positive curvature. Normalized flow on G_2 results in convergence to positive constant curvature with non zero edge weights consisting of edges connecting degree $n+1$ and degree n nodes, as well as the one pendant edge. The results suggest that normalized flow results in a positive constant curvature with limiting metric depending on the size of n.



G_3



The results of G_3 show the behavior of a graph where 3 edge types are identifiable by induced symmetry with two pendant edges. The results suggest that unnormalized flow results in all edge weights decaying to weight 0, while normalized flow results in the pendant leaf edges decaying to weight 0 while the three internal edges converge to a positive limiting weight with constant 0 curvature. This graph structure is not covered by the propositions in this paper, but has a similar symmetry induced structure which results in convergence.

Conclusion

In this paper, we examined four different special graph structures and analyzed the long time behavior of the Ollivier-Lin-Lu-Yau version of Discrete Ricci Flow on the graphs. We presented four propositions: **Proposition 1** shows convergence of a Level-2 star graph, **Proposition 2** and **Proposition 3** show convergence of two special cyclic graph structures, and **Proposition 4** shows a partial convergence result of a third special cyclic graph structure. The propositions and experimental results suggest that graphs with highly symmetrical structures converge to limiting metrics under Discrete Ricci Flow.

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